

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

```
In [2]: df = pd.read_csv("loan_data_set.csv")
```

```
In [4]: df.head()
```

```
Out[4]:
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIn
0	LP001002	Male	No	0	Graduate	No	
1	LP001003	Male	Yes	1	Graduate	No	
2	LP001005	Male	Yes	0	Graduate	Yes	
3	LP001006	Male	Yes	0	Not Graduate	No	
4	LP001008	Male	No	0	Graduate	No	

```
In [5]: df.shape
```

```
Out[5]: (614, 13)
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 614 entries, 0 to 613
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID               614 non-null   object
1   Gender                601 non-null   object
2   Married               611 non-null   object
3   Dependents            599 non-null   object
4   Education             614 non-null   object
5   Self_Employed         582 non-null   object
6   ApplicantIncome       614 non-null   int64
7   CoapplicantIncome     614 non-null   float64
8   LoanAmount            592 non-null   float64
9   Loan_Amount_Term      600 non-null   float64
10  Credit_History        564 non-null   float64
11  Property_Area         614 non-null   object
12  Loan_Status           614 non-null   object
dtypes: float64(4), int64(1), object(8)
memory usage: 62.5+ KB
```

```
In [7]: df.describe()
```

Out[7]:

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
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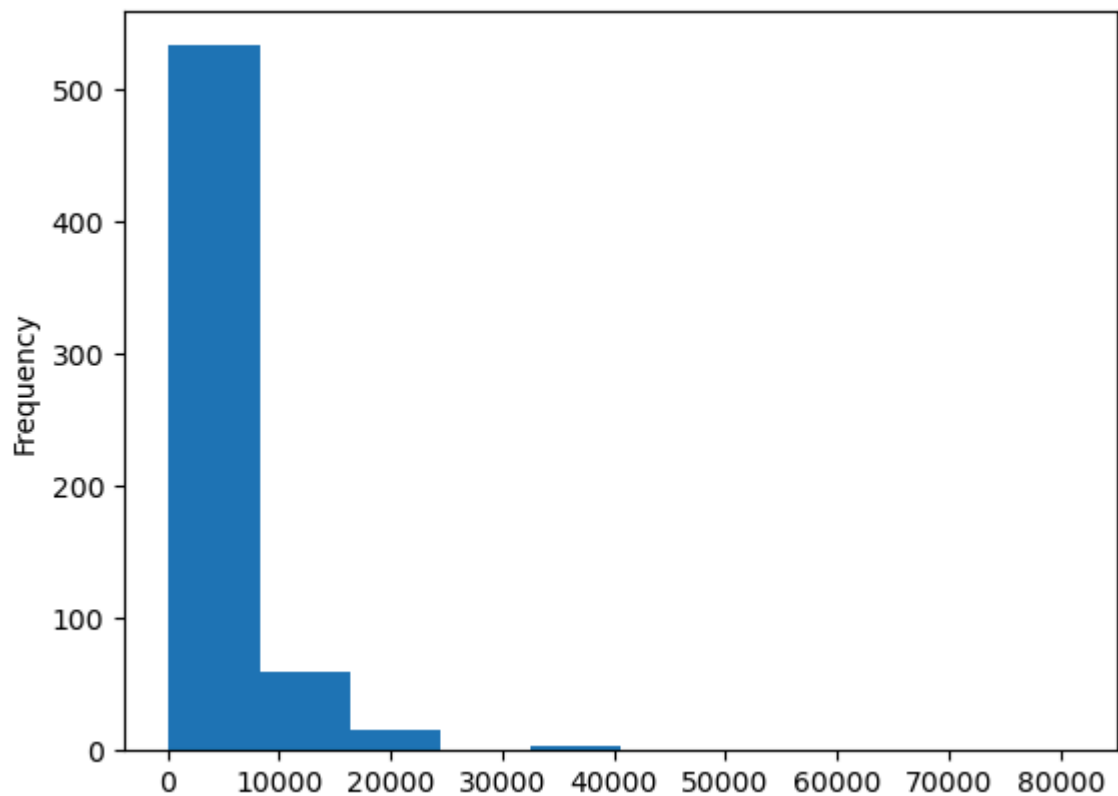
count	614.000000	614.000000	592.000000	600.000000	614.000000
mean	5403.459283	1621.245798	146.412162	342.000000	0.988761
std	6109.041673	2926.248369	85.587325	65.12041	0.191238
min	150.000000	0.000000	9.000000	12.000000	0.000000
25%	2877.500000	0.000000	100.000000	360.000000	0.000000
50%	3812.500000	1188.500000	128.000000	360.000000	0.000000
75%	5795.000000	2297.250000	168.000000	360.000000	0.000000
max	81000.000000	41667.000000	700.000000	480.000000	1.000000

In [8]: `df.isna().sum()`

Out[8]:

Loan_ID	0
Gender	13
Married	3
Dependents	15
Education	0
Self_Employed	32
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	22
Loan_Amount_Term	14
Credit_History	50
Property_Area	0
Loan_Status	0
dtype:	int64

In [9]: `df["ApplicantIncome"].plot(kind="hist")`
`plt.show()`



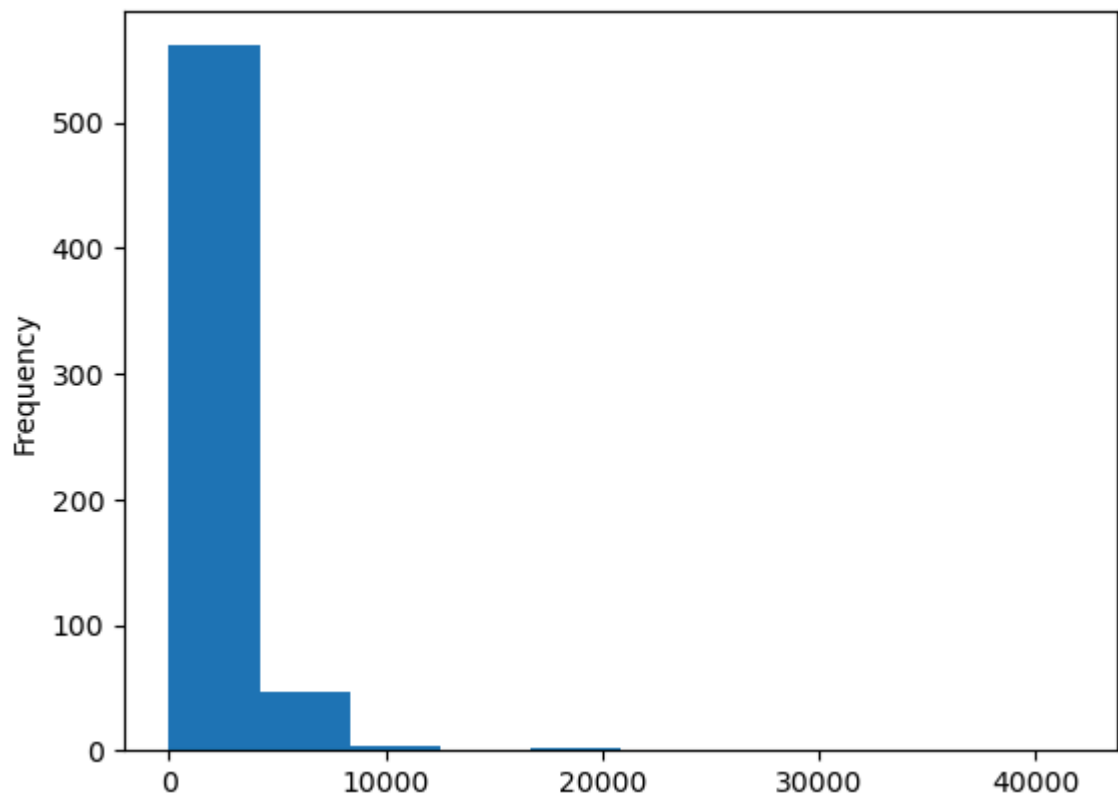
```
In [10]: df["ApplicantIncome"].fillna(df["ApplicantIncome"].mean(), inplace=True)
df
```

```
Out[10]:
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome
0	LP001002	Male	No	0	Graduate	No	1195
1	LP001003	Male	Yes	1	Graduate	No	1360
2	LP001005	Male	Yes	0	Graduate	Yes	9800
3	LP001006	Male	Yes	0	Not Graduate	No	1305
4	LP001008	Male	No	0	Graduate	No	2200
...	...	...	...	...	...	...	...
609	LP002978	Female	No	0	Graduate	No	1305
610	LP002979	Male	Yes	3+	Graduate	No	2013
611	LP002983	Male	Yes	1	Graduate	No	1305
612	LP002984	Male	Yes	2	Graduate	No	1305
613	LP002990	Female	No	0	Graduate	Yes	9800

614 rows × 13 columns

```
In [11]: df["CoapplicantIncome"].plot(kind="hist")
plt.show()
```



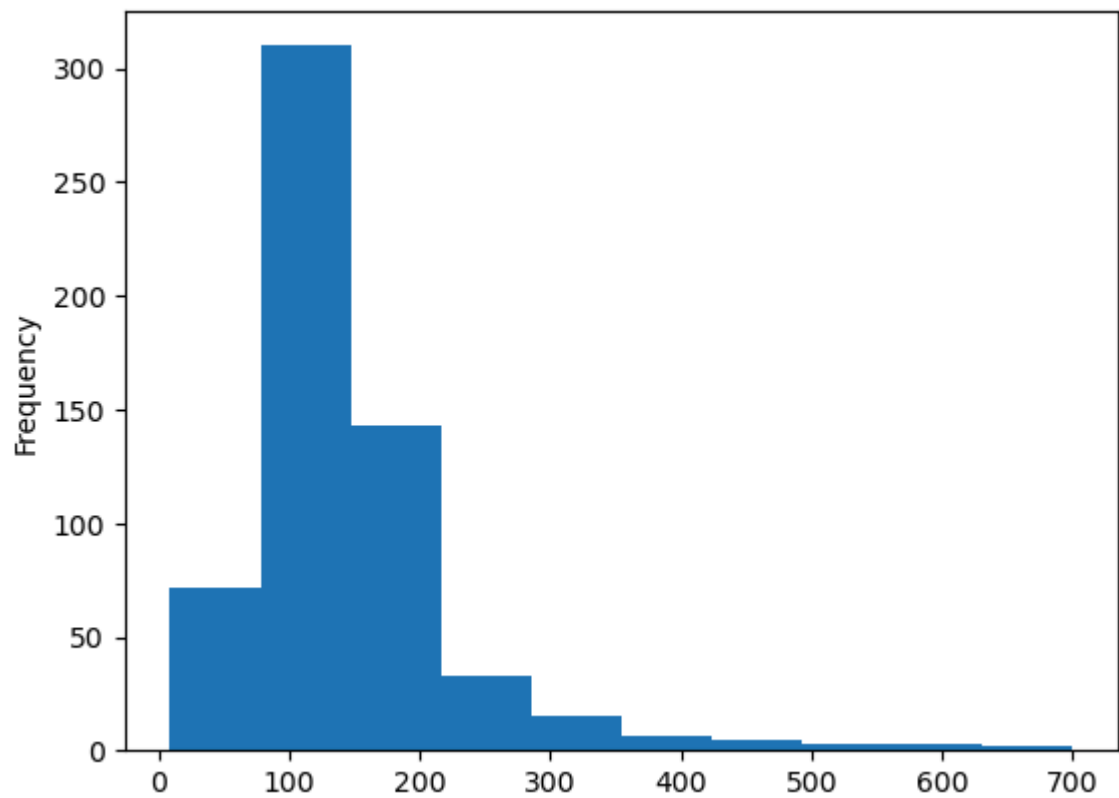
```
In [12]: df["CoapplicantIncome"].fillna(df["CoapplicantIncome"].mean(), inplace=True)
df
```

```
Out[12]:
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	Applicant
0	LP001002	Male	No	0	Graduate	No	
1	LP001003	Male	Yes	1	Graduate	No	
2	LP001005	Male	Yes	0	Graduate	Yes	
3	LP001006	Male	Yes	0	Not Graduate	No	
4	LP001008	Male	No	0	Graduate	No	
...	...	...	...	...	...	...	...
609	LP002978	Female	No	0	Graduate	No	
610	LP002979	Male	Yes	3+	Graduate	No	
611	LP002983	Male	Yes	1	Graduate	No	
612	LP002984	Male	Yes	2	Graduate	No	
613	LP002990	Female	No	0	Graduate	Yes	

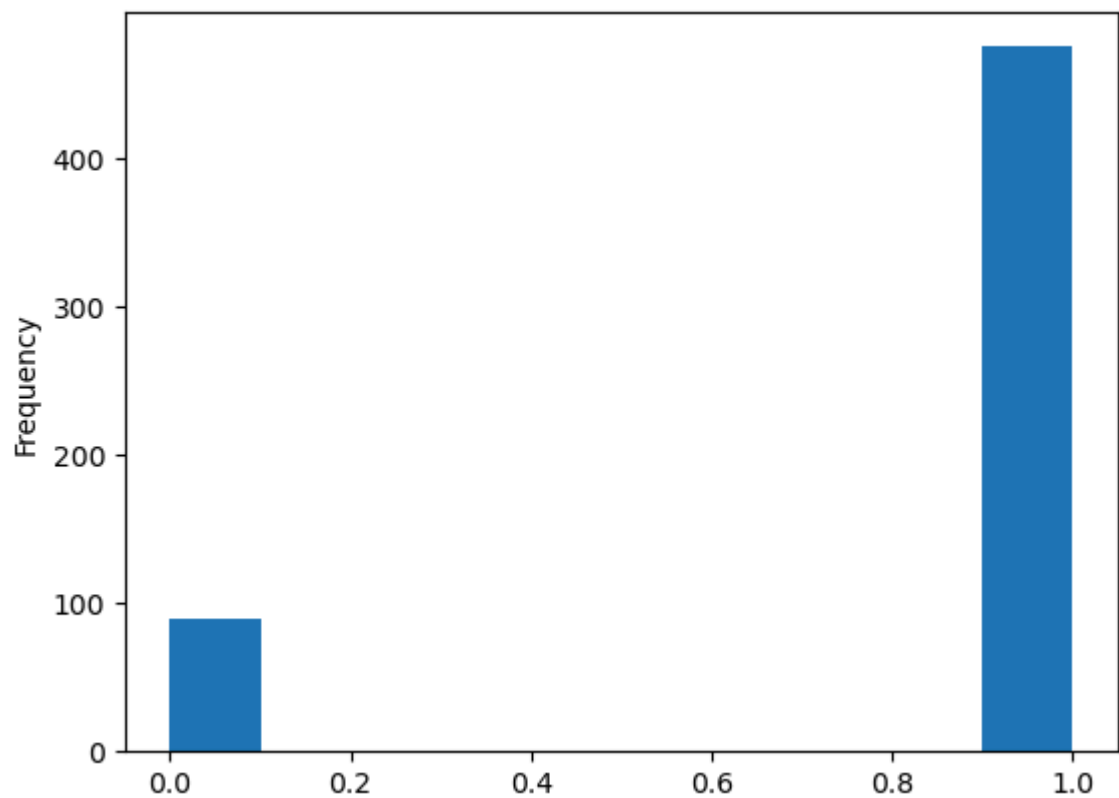
614 rows × 13 columns

```
In [13]: df["LoanAmount"].plot(kind="hist")
plt.show()
```



```
In [14]: df["LoanAmount"].fillna(df["LoanAmount"].mean(), inplace=True)
```

```
In [15]: df["Credit_History"].plot(kind="hist")  
plt.show()
```



```
In [16]: df["Credit_History"].fillna(np.random.randint(0,2), inplace=True)  
df
```

```
Out[16]:
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	Applicant
0	LP001002	Male	No	0	Graduate	No	
1	LP001003	Male	Yes	1	Graduate	No	
2	LP001005	Male	Yes	0	Graduate	Yes	
3	LP001006	Male	Yes	0	Not Graduate	No	
4	LP001008	Male	No	0	Graduate	No	
...	...	...	...	...	...	...	...
609	LP002978	Female	No	0	Graduate	No	
610	LP002979	Male	Yes	3+	Graduate	No	
611	LP002983	Male	Yes	1	Graduate	No	
612	LP002984	Male	Yes	2	Graduate	No	
613	LP002990	Female	No	0	Graduate	Yes	

614 rows × 13 columns

```
In [17]: grouped_df = df[["ApplicantIncome", "CoapplicantIncome", "LoanAmount", "C
```

```
In [18]: mean = grouped_df.mean()
mean
```

```
Out[18]:
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Credit_History
Loan_Status				
N	5446.078125	1877.807292	150.945488	0.505208
Y	5384.068720	1504.516398	144.349606	0.895735

```
In [19]: median = grouped_df.median()
median
```

```
Out[19]:
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Credit_History
Loan_Status				
N	3833.5	268.0	133.5	1.0
Y	3812.5	1239.5	128.0	1.0

```
In [20]: min = grouped_df.min()
min
```

```
Out[20]:
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Credit_History
Loan_Status				
N	150	0.0	9.0	0.0
Y	210	0.0	17.0	0.0

```
In [21]: max = grouped_df.max()
max
```

```
Out[21]:
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Credit_History
Loan_Status				
N	81000	41667.0	570.0	1.0
Y	63337	20000.0	700.0	1.0

```
In [22]: std = grouped_df.std()
std
```

```
Out[22]:
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Credit_History
Loan_Status				
N	6819.558528	4384.060103	83.361163	0.501280
Y	5765.441615	1924.754855	84.361109	0.305967

```
In [24]: data = pd.read_csv("Iris.csv")
data.keys()
```

```
Out[24]: Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
               'Species'],
              dtype='object')
```

```
In [25]: data.shape
```

```
Out[25]: (150, 6)
```

```
In [26]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Id              150 non-null   int64
1   SepalLengthCm   150 non-null   float64
2   SepalWidthCm    150 non-null   float64
3   PetalLengthCm   150 non-null   float64
4   PetalWidthCm    150 non-null   float64
5   Species         150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
In [27]: data.describe()
```

```
Out[27]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
In [28]: data["SepalLengthCm"].describe()
```

```
Out[28]:
```

count	150.000000
mean	5.843333
std	0.828066
min	4.300000
25%	5.100000
50%	5.800000
75%	6.400000
max	7.900000

Name: SepalLengthCm, dtype: float64

```
In [29]: data.groupby("Species").describe()
```

```
Out[29]:
```

									Id	SepalLength
	count	mean	std	min	25%	50%	75%	max	count	m
Species										
Iris-setosa	50.0	25.5	14.57738	1.0	13.25	25.5	37.75	50.0	50.0	5.
Iris-versicolor	50.0	75.5	14.57738	51.0	63.25	75.5	87.75	100.0	50.0	5.
Iris-virginica	50.0	125.5	14.57738	101.0	113.25	125.5	137.75	150.0	50.0	6.

3 rows × 40 columns

```
In [30]: data.groupby("Species").describe().sum()
```



```

Out[30]: Id          count    150.000000
          mean      226.500000
          std       43.732139
          min       153.000000
          25%       189.750000
          50%       226.500000
          75%       263.250000
          max       300.000000
SepalLengthCm count    150.000000
          mean      17.530000
          std       1.504540
          min       14.100000
          25%       16.625000
          50%       17.400000
          75%       18.400000
          max       20.700000
SepalWidthCm  count    150.000000
          mean       9.162000
          std       1.017319
          min       6.500000
          25%       8.450000
          50%       9.200000
          75%       9.850000
          max       11.600000
PetalLengthCm count    150.000000
          mean      11.276000
          std       1.195317
          min       8.500000
          25%      10.500000
          50%      11.400000
          75%      12.050000
          max      13.900000
PetalWidthCm  count    150.000000
          mean       3.596000
          std       0.579612
          min       2.500000
          25%       3.200000
          50%       3.500000
          75%       4.100000
          max       4.900000
dtype: float64

```

```

In [31]: data.groupby(["Species"])[ "SepalLengthCm"].mean()

```

```

Out[31]: Species
Iris-setosa      5.006
Iris-versicolor  5.936
Iris-virginica   6.588
Name: SepalLengthCm, dtype: float64

```

```

In [32]: data.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Id              150 non-null    int64
 1   SepalLengthCm   150 non-null    float64
 2   SepalWidthCm    150 non-null    float64
 3   PetalLengthCm   150 non-null    float64
 4   PetalWidthCm    150 non-null    float64
 5   Species         150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB

```

```
In [33]: data['Species'].value_counts()
```

```

Out[33]: Species
Iris-setosa      50
Iris-versicolor  50
Iris-virginica   50
Name: count, dtype: int64

```

```
In [34]: data.Species.mode()
```

```

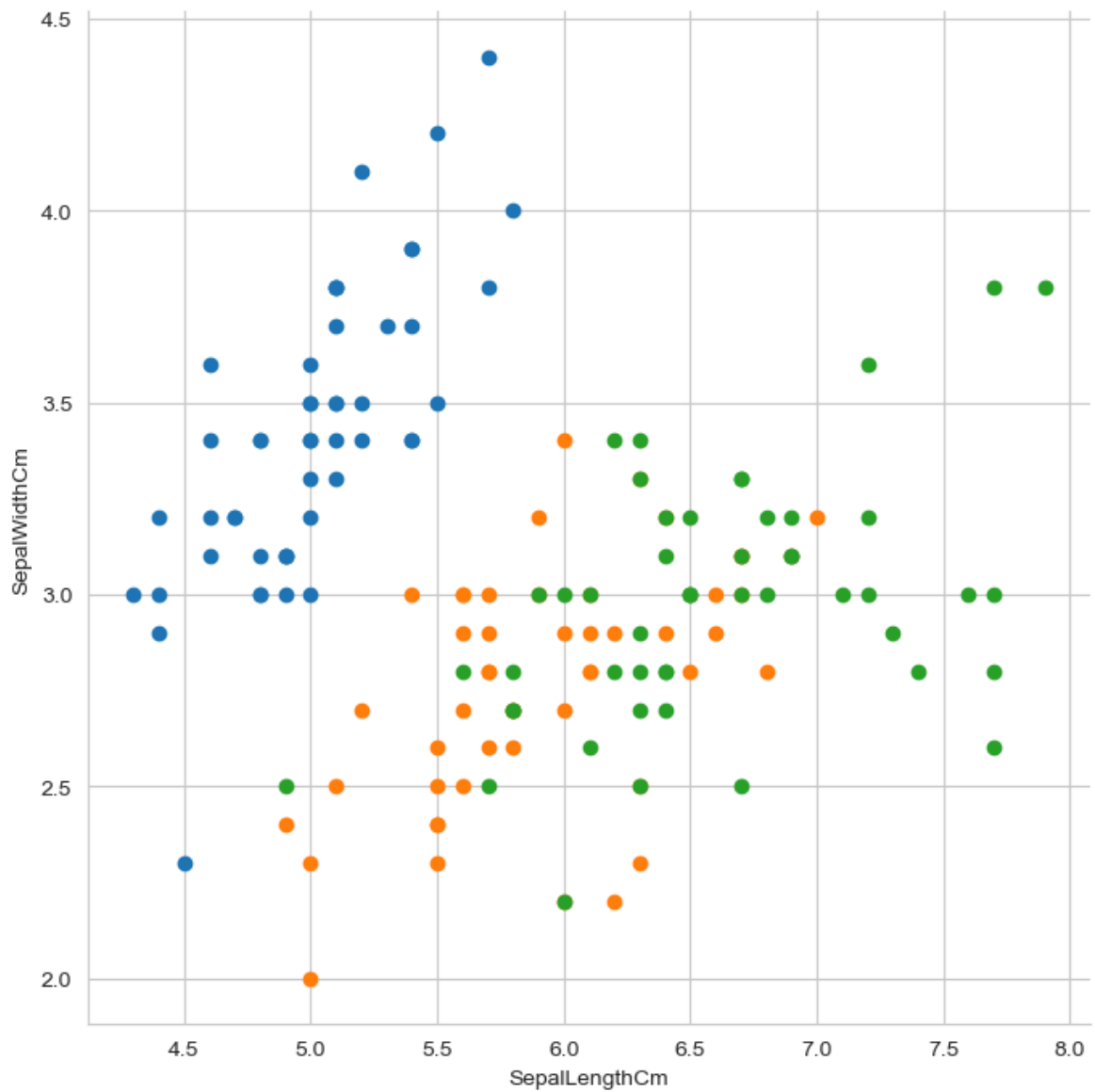
Out[34]: 0      Iris-setosa
         1      Iris-versicolor
         2      Iris-virginica
         Name: Species, dtype: object

```

```

In [35]: import seaborn as sns
sns.set_style("whitegrid")
sns.FacetGrid(data, hue="Species", height=7).map(plt.scatter, "SepalLength", "PetalLength")
plt.show()

```



```
In [36]: setosa_data = data[data['Species'] == 'Iris-setosa']  
versicolor_data = data[data['Species'] == 'Iris-versicolor']  
virginica_data = data[data['Species'] == 'Iris-virginica']
```

```
In [37]: setosa_data
```

Out[37]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
5	6	5.4	3.9	1.7	0.4	Iris-setosa
6	7	4.6	3.4	1.4	0.3	Iris-setosa
7	8	5.0	3.4	1.5	0.2	Iris-setosa
8	9	4.4	2.9	1.4	0.2	Iris-setosa
9	10	4.9	3.1	1.5	0.1	Iris-setosa
10	11	5.4	3.7	1.5	0.2	Iris-setosa
11	12	4.8	3.4	1.6	0.2	Iris-setosa
12	13	4.8	3.0	1.4	0.1	Iris-setosa
13	14	4.3	3.0	1.1	0.1	Iris-setosa
14	15	5.8	4.0	1.2	0.2	Iris-setosa
15	16	5.7	4.4	1.5	0.4	Iris-setosa
16	17	5.4	3.9	1.3	0.4	Iris-setosa
17	18	5.1	3.5	1.4	0.3	Iris-setosa
18	19	5.7	3.8	1.7	0.3	Iris-setosa
19	20	5.1	3.8	1.5	0.3	Iris-setosa

Id		SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
20	21	5.4	3.4	1.7	0.2	Iris-setosa
21	22	5.1	3.7	1.5	0.4	Iris-setosa
22	23	4.6	3.6	1.0	0.2	Iris-setosa
23	24	5.1	3.3	1.7	0.5	Iris-setosa
24	25	4.8	3.4	1.9	0.2	Iris-setosa
25	26	5.0	3.0	1.6	0.2	Iris-setosa
26	27	5.0	3.4	1.6	0.4	Iris-setosa
27	28	5.2	3.5	1.5	0.2	Iris-setosa
28	29	5.2	3.4	1.4	0.2	Iris-setosa
29	30	4.7	3.2	1.6	0.2	Iris-setosa
30	31	4.8	3.1	1.6	0.2	Iris-setosa
31	32	5.4	3.4	1.5	0.4	Iris-setosa
32	33	5.2	4.1	1.5	0.1	Iris-setosa
33	34	5.5	4.2	1.4	0.2	Iris-setosa
34	35	4.9	3.1	1.5	0.1	Iris-setosa
35	36	5.0	3.2	1.2	0.2	Iris-setosa
36	37	5.5	3.5	1.3	0.2	Iris-setosa
37	38	4.9	3.1	1.5	0.1	Iris-setosa
38	39	4.4	3.0	1.3	0.2	Iris-setosa
39	40	5.1	3.4	1.5	0.2	Iris-setosa

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
40	41	5.0	3.5	1.3	0.3	Iris-setosa
41	42	4.5	2.3	1.3	0.3	Iris-setosa
42	43	4.4	3.2	1.3	0.2	Iris-setosa
43	44	5.0	3.5	1.6	0.6	Iris-setosa
44	45	5.1	3.8	1.9	0.4	Iris-setosa
45	46	4.8	3.0	1.4	0.3	Iris-setosa
46	47	5.1	3.8	1.6	0.2	Iris-setosa
47	48	4.6	3.2	1.4	0.2	Iris-setosa
48	49	5.3	3.7	1.5	0.2	Iris-setosa
49	50	5.0	3.3	1.4	0.2	Iris-setosa

In [38]: versicolor_data

Out[38]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
50	51	7.0	3.2	4.7	1.4	Iris-versicolor
51	52	6.4	3.2	4.5	1.5	Iris-versicolor
52	53	6.9	3.1	4.9	1.5	Iris-versicolor
53	54	5.5	2.3	4.0	1.3	Iris-versicolor
54	55	6.5	2.8	4.6	1.5	Iris-versicolor
55	56	5.7	2.8	4.5	1.3	Iris-versicolor
56	57	6.3	3.3	4.7	1.6	Iris-versicolor
57	58	4.9	2.4	3.3	1.0	Iris-versicolor
58	59	6.6	2.9	4.6	1.3	Iris-versicolor
59	60	5.2	2.7	3.9	1.4	Iris-versicolor
60	61	5.0	2.0	3.5	1.0	Iris-versicolor
61	62	5.9	3.0	4.2	1.5	Iris-versicolor
62	63	6.0	2.2	4.0	1.0	Iris-versicolor
63	64	6.1	2.9	4.7	1.4	Iris-versicolor
64	65	5.6	2.9	3.6	1.3	Iris-versicolor
65	66	6.7	3.1	4.4	1.4	Iris-versicolor
66	67	5.6	3.0	4.5	1.5	Iris-versicolor
67	68	5.8	2.7	4.1	1.0	Iris-versicolor
68	69	6.2	2.2	4.5	1.5	Iris-versicolor
69	70	5.6	2.5	3.9	1.1	Iris-versicolor

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
70	71	5.9	3.2	4.8	1.8	Iris-versicolor
71	72	6.1	2.8	4.0	1.3	Iris-versicolor
72	73	6.3	2.5	4.9	1.5	Iris-versicolor
73	74	6.1	2.8	4.7	1.2	Iris-versicolor
74	75	6.4	2.9	4.3	1.3	Iris-versicolor
75	76	6.6	3.0	4.4	1.4	Iris-versicolor
76	77	6.8	2.8	4.8	1.4	Iris-versicolor
77	78	6.7	3.0	5.0	1.7	Iris-versicolor
78	79	6.0	2.9	4.5	1.5	Iris-versicolor
79	80	5.7	2.6	3.5	1.0	Iris-versicolor
80	81	5.5	2.4	3.8	1.1	Iris-versicolor
81	82	5.5	2.4	3.7	1.0	Iris-versicolor
82	83	5.8	2.7	3.9	1.2	Iris-versicolor
83	84	6.0	2.7	5.1	1.6	Iris-versicolor
84	85	5.4	3.0	4.5	1.5	Iris-versicolor
85	86	6.0	3.4	4.5	1.6	Iris-versicolor
86	87	6.7	3.1	4.7	1.5	Iris-versicolor
87	88	6.3	2.3	4.4	1.3	Iris-versicolor
88	89	5.6	3.0	4.1	1.3	Iris-versicolor
89	90	5.5	2.5	4.0	1.3	Iris-versicolor

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
90	91	5.5	2.6	4.4	1.2	Iris- versicolor
91	92	6.1	3.0	4.6	1.4	Iris- versicolor
92	93	5.8	2.6	4.0	1.2	Iris- versicolor
93	94	5.0	2.3	3.3	1.0	Iris- versicolor
94	95	5.6	2.7	4.2	1.3	Iris- versicolor
95	96	5.7	3.0	4.2	1.2	Iris- versicolor
96	97	5.7	2.9	4.2	1.3	Iris- versicolor
97	98	6.2	2.9	4.3	1.3	Iris- versicolor
98	99	5.1	2.5	3.0	1.1	Iris- versicolor
99	100	5.7	2.8	4.1	1.3	Iris- versicolor

In [39]: virginica_data

Out[39]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
100	101	6.3	3.3	6.0	2.5	Iris-virginica
101	102	5.8	2.7	5.1	1.9	Iris-virginica
102	103	7.1	3.0	5.9	2.1	Iris-virginica
103	104	6.3	2.9	5.6	1.8	Iris-virginica
104	105	6.5	3.0	5.8	2.2	Iris-virginica
105	106	7.6	3.0	6.6	2.1	Iris-virginica
106	107	4.9	2.5	4.5	1.7	Iris-virginica
107	108	7.3	2.9	6.3	1.8	Iris-virginica
108	109	6.7	2.5	5.8	1.8	Iris-virginica
109	110	7.2	3.6	6.1	2.5	Iris-virginica
110	111	6.5	3.2	5.1	2.0	Iris-virginica
111	112	6.4	2.7	5.3	1.9	Iris-virginica
112	113	6.8	3.0	5.5	2.1	Iris-virginica
113	114	5.7	2.5	5.0	2.0	Iris-virginica
114	115	5.8	2.8	5.1	2.4	Iris-virginica
115	116	6.4	3.2	5.3	2.3	Iris-virginica
116	117	6.5	3.0	5.5	1.8	Iris-virginica
117	118	7.7	3.8	6.7	2.2	Iris-virginica
118	119	7.7	2.6	6.9	2.3	Iris-virginica
119	120	6.0	2.2	5.0	1.5	Iris-virginica

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
120	121	6.9	3.2	5.7	2.3	Iris-virginica
121	122	5.6	2.8	4.9	2.0	Iris-virginica
122	123	7.7	2.8	6.7	2.0	Iris-virginica
123	124	6.3	2.7	4.9	1.8	Iris-virginica
124	125	6.7	3.3	5.7	2.1	Iris-virginica
125	126	7.2	3.2	6.0	1.8	Iris-virginica
126	127	6.2	2.8	4.8	1.8	Iris-virginica
127	128	6.1	3.0	4.9	1.8	Iris-virginica
128	129	6.4	2.8	5.6	2.1	Iris-virginica
129	130	7.2	3.0	5.8	1.6	Iris-virginica
130	131	7.4	2.8	6.1	1.9	Iris-virginica
131	132	7.9	3.8	6.4	2.0	Iris-virginica
132	133	6.4	2.8	5.6	2.2	Iris-virginica
133	134	6.3	2.8	5.1	1.5	Iris-virginica
134	135	6.1	2.6	5.6	1.4	Iris-virginica
135	136	7.7	3.0	6.1	2.3	Iris-virginica
136	137	6.3	3.4	5.6	2.4	Iris-virginica
137	138	6.4	3.1	5.5	1.8	Iris-virginica
138	139	6.0	3.0	4.8	1.8	Iris-virginica
139	140	6.9	3.1	5.4	2.1	Iris-virginica

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
140	141	6.7	3.1	5.6	2.4	Iris-virginica
141	142	6.9	3.1	5.1	2.3	Iris-virginica
142	143	5.8	2.7	5.1	1.9	Iris-virginica
143	144	6.8	3.2	5.9	2.3	Iris-virginica
144	145	6.7	3.3	5.7	2.5	Iris-virginica
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

```
In [42]: # Setosa
setosa_stats = {
    'percentile': np.percentile(setosa_data['SepalLengthCm'], 50),
    'mean': np.mean(setosa_data['SepalLengthCm']),
    'std_dev': np.std(setosa_data['SepalLengthCm'])
}

# Versicolor
versicolor_stats = {
    'percentile': np.percentile(versicolor_data['SepalLengthCm'], 50),
    'mean': np.mean(versicolor_data['SepalLengthCm']),
    'std_dev': np.std(versicolor_data['SepalLengthCm'])
}

# Virginica
virginica_stats = {
    'percentile': np.percentile(virginica_data['SepalLengthCm'], 50),
    'mean': np.mean(virginica_data['SepalLengthCm']),
    'std_dev': np.std(virginica_data['SepalLengthCm'])
}

# Print results
print("Statistics for Iris Setosa:")
print(setosa_stats)

print("\nStatistics for Iris Versicolor:")
print(versicolor_stats)
```

```
print("\nStatistics for Iris Virginica:")  
print(virginica_stats)
```

Statistics for Iris Setosa:

```
{'percentile': np.float64(5.0), 'mean': np.float64(5.006), 'std_dev': 0.34  
8946987377739}
```

Statistics for Iris Versicolor:

```
{'percentile': np.float64(5.9), 'mean': np.float64(5.936), 'std_dev': 0.51  
09833656783752}
```

Statistics for Iris Virginica:

```
{'percentile': np.float64(6.5), 'mean': np.float64(6.587999999999998), 'st  
d_dev': 0.6294886813914925}
```

In []: